

Trust Issues with Trust Scales: Examining the Psychometric Quality of Trust Measures in the Context of AI

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ABSTRACT

Trust is crucial for human interaction with artificial intelligence (AI) and is frequently measured through questionnaires or rating scales. One commonly used questionnaire in AI research is the *Trust between People and Automation* scale (TPA). However, its psychometric quality has yet to be examined in the context of AI. More recently, a *Trust Scale for Explainable AI* (TXAI) was recommended but not empirically evaluated. In this study, we assessed the psychometric qualities of both scales, using confirmatory and exploratory factor analyses to test the scales' validity and coefficients α and ω for reliability estimation. Our results suggested good psychometric quality for the TXAI after removing one item. Concerning the TPA, acceptable quality was only achieved when using a two-factor model (trust and distrust) and after removing two items. We provide recommendations for using the two scales and evidence to distinguish trust and distrust as separate psychological constructs.

CCS CONCEPTS

• Human-centered computing → HCI theory, concepts and models; Empirical studies in HCI; User studies.

KEYWORDS

AI, XAI, Trust, Measurement, Questionnaires, Validation, Psychometrics

ACM Reference Format:

Sebastian A. C. Perrig, Nicolas Scharowski, and Florian Brühlmann. 2023. Trust Issues with Trust Scales: Examining the Psychometric Quality of Trust Measures in the Context of AI. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585808>

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585808>

1 INTRODUCTION

With advances in artificial intelligence (AI), there is an increasing need for understanding and measuring human trust in AI as it plays a key role in people's reliance on AI systems [12, 20]. Lee and See defined *trust* as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability." [20, p. 6]. The definition of trust as an *attitude* implies that it is a psychological construct that is not directly observable [14] and it should therefore be distinguished from behavioral measures such as reliance [30]. Research often measures such unobservable constructs through questionnaires or rating scales. At their core, questionnaires are about using a series of items to measure a construct of interest [6]. For the construct of "trust," researchers may let participants fill out a questionnaire and rate multiple items on a scale (e.g., a Likert-type scale ranging from 1 - 7). After analyzing the responses to these items, a number is calculated that reflects the participants' subjective level of trust, which can then be used to make statistical inferences.

Buçinca et al. [3] argued that measuring trust via questionnaires has been a focal point in AI research. It is common for researchers to use single items [e.g., 40] or self-developed questionnaires [e.g., 25, 39]. However, using self-developed questionnaires can be problematic as they usually lack a rigid construction and quality assurance process [8] and are often only used in a single study, making comparability between different studies difficult. For this reason, it is better to use standardized questionnaires whose psychometric quality (i.e., objectivity, reliability, and validity) has been scrutinized. However, as there is no dedicated "trust in AI" scale, researchers have used scales from AI-related fields, such as the *Trust between People and Automation* scale (TPA) by Jian et al. [17], which is a frequently used scale in the evaluation of Human-AI Interaction [37]. However, using scales from other fields requires adapting the questionnaires rather than using them as intended. As the name suggests, the TPA was designed to study *trust in automation* more generally rather than *trust in AI* specifically. Because of this, AI researchers often need to adapt and rephrase the TPA items, for example, "the system is dependable" to "the machine learning system is dependable." Terminological differences can affect people's perceptions and evaluations of technology [19]. This raises the question whether an adapted scale still measures what it was initially intended to measure, and its psychometric quality must be reassessed [8, 18]. More recently, Hoffman et al. [13] recommended

a trust scale in the context of XAI (from here on called TXAI) that is based on and adapted from existing trust scales, including the TPA by Jian et al. [17]. While adopting items from existing scales is a common first step when developing new questionnaires, psychometric quality assessment must follow to ensure its validity and reliability [8]. To the best of our knowledge, the TXAI has never been empirically investigated, while the TPA's psychometric quality still needs to be rigorously examined in an AI context.

Past research [9, 35] that validated the TPA scale in contexts other than AI has provided valuable insights. For example, in the original publication by Jian et al. and in line with Mayer et al. [23], the authors thought of trust as a single-dimensional construct, with trust and distrust forming two ends of a continuum. However, Spain et al. [35] showed that items for trust and distrust load on two separate factors, supporting Lewicki et al. [21], who proposed trust and distrust as two distinct constructs. Researchers typically conceptualize and measure attitudes as lying along a bipolar continuum ranging from negative (e.g., distrust) to positive (e.g., trust) [7], but attitudes can sometimes possess both positive and negative features [28]. For example, smokers trying to quit smoking may have both positive and negative feelings towards cigarettes, suggesting that positive and negative attitudes can coexist simultaneously [4]. Concerning the relationship between trust and distrust in the context of AI, the debate is still ongoing [31].

The contribution of this paper is threefold. First, we present an initial psychometric evaluation of Jian et al.'s TPA scale in the context of AI. Second, we investigate a two-factor structure for the TPA as proposed outside of an AI context by Spain et al. [35], comparing it to the originally proposed single-factor model. Third, we report on a first and independent psychometric evaluation of the TXAI by Hoffman et al. Finally, we compare the two trust scales and derive recommendations for researchers who aim to measure trust within an AI context. In line with this, our findings enable future researchers to use these scales more accurately and confidently.

2 METHOD

We collected ratings on the TPA and TXAI as part of a larger study examining the impact of human-centered explanations of AI recommendations. Following past work [27], the study was conducted in the real estate valuation domain, where machine learning is often used to predict apartment prices. Airbnb and Zillow are examples of real estate websites that provide price recommendations in this way. Participants were presented with a scenario in which they had to estimate subleasing prices for six apartments, which were real listings from the website Zillow in May 2020. The features and amenities of the apartments were directly collected from the listings. All participants were presented with the same apartment listings and features, but the price recommendations provided to them differed.

2.1 Procedure

After providing informed consent, participants were asked to imagine they were trying to sublease apartments on a website. They were provided information about the features and amenities of each apartment (e.g., number of bedrooms, distance to public transit) and were instructed to estimate an initial subleasing price. After this

initial estimation, participants received a recommendation from an algorithm introduced as an AI system. Participants then had the opportunity to adjust their subleasing price, closely following the procedure by Poursabzi-Sangdeh et al. [27]. After completing the task, participants filled out the two trust questionnaires, TPA and TXAI, and provided demographic information.

To create an element of uncertainty, which is a crucial aspect of trust [23], participants were told that they would be reimbursed based on their performance. They were informed that the top 10% would receive a bonus for good performance and that the bottom 10% of performance would be deemed insufficient and their work would be rejected. In reality, all participants with sufficient data quality received an additional \$1.00 bonus, and they were debriefed about this procedure immediately after the study. The participants' uncertainty was assessed using a single item, with a seven-point Likert-type response scale ranging from 1 ("not at all severe") to 7 ("extremely severe"). Participants rated the prospect of not getting paid due to poor performance as moderately severe ($M = 4.91$, $SD = 1.84$).

2.2 Measures

The TXAI by Hoffman et al. uses a five-point Likert-type scale ranging from 1 ("I disagree strongly") to 5 ("I agree strongly") and consists of eight items, of which one is negatively formulated. The TPA scale by Jian et al. consists of twelve items, five of which are reverse-coded (rephrased from positive to negative meaning). Answers are collected using a seven-point Likert-type scale ranging from 1 ("not at all") to 7 ("extremely"). Both scales were used in their original form, except for prefixing the word "AI-" with the word "system" for the TPA scale (e.g., "I have confidence in the AI-system") and substituting the word "tool" with "AI" for the TXAI (e.g., "The AI is very reliable"). The reverse-coded/negatively formulated items for both scales were re-coded prior to data analysis.

2.3 Participants

We recruited 2373 participants on Amazon Mechanical Turk. Only workers from the United States with a 95% approval rating on human-intelligence-tasks and at least 100 approved tasks were allowed to participate. Those who completed the task received \$1.50 and a bonus of \$1.00 as reimbursement. Following recommendations by Brühlmann et al. [2], the data was cleaned using several criteria, including removing participants who failed to correctly answer control questions or self-declared not to use their data. After applying these criteria, 1368 participants were included in the final data analysis. Most of these participants identified as men (52%) and had an average age of 40 years ($M = 40.41$, $SD = 12.15$, $min = 18$, $max = 89$).

3 RESULTS

The following results highlight key findings from the psychometric quality investigation of the two trust scales. Results were obtained using the statistical software R (version 4.2.2). We refer readers to OSF for the data and complete analysis script (<https://osf.io/5vgk8/>).

Table 1: Fit indices for CFA models of the trust scales.

Model	χ^2	df	p-value	χ^2	RMSEA	SRMR	CFI
TXAI [13]: one factor	85.81	20	< .001	.056	.032	.983	
TXAI [13]: one factor, item 6 removed	32.07	14	.004	.036	.017	.995	
TXAI [13]: one factor, items 2, 5, 6 & 7 removed	1.17	2	.557	.000	.004	1.000	
TPA [17]: one factor	4657.62	54	< .001	.286	.283	.426	
TPA [17]: two factors	520.39	53	< .001	.094	.097	.939	
TPA [17]: two factors, items 8 & 12 removed	284.82	34	< .001	.089	.079	.961	

Note: Robust values are reported wherever possible.

3.1 CFA TXAI - Original Model

We started by investigating a single-factor structure for the TXAI items using confirmatory factor analysis (CFA). We considered recommendations by Hu and Bentler [16] as indication of good model fit.¹ Multivariate normality of the data was not given, tested with both Henze-Zirkler test [11] and Mardia's test [22], leading us to use a robust maximum likelihood estimator. Fit indices for this and all other CFAs we conducted are presented in Table 1. Results from the single-factor CFA indicated a good model fit based on *RMSEA*, *SRMR*, and *CFI*, but not based on the χ^2 test, which is not surprising given that the χ^2 test is affected by sample size and deviations from multivariate normality [34]. Nevertheless, we decided to conduct exploratory factor analyses (EFAs) to search for alternative models that fit the data better.

3.2 EFA TXAI

We followed the .40 – .30 – .20 rule [15] to interpret EFA factor loadings. For the interpretation of communalities, values < .50 were considered problematic [10]. Given the violation of multivariate normality, we worked with a principal axis factoring extraction method for all EFAs. We further used an oblique rotation method, expecting correlations among the factors. Assumptions for EFA were met: significant Bartlett's test for sphericity ($\chi^2(28) = 4938.38$, $p < .001$) and an adequate Kaiser-Meyer-Olkin test (all $\geq .90$). To determine the number of factors, we used parallel analysis and a scree-plot, which suggested a two-factor model. The two-factor solution for the TXAI explained 52% of variance, although most of it was explained by the first factor (48%), while the second factor only explained 5%. The two factors correlate at .20. Items all loaded well onto a single factor, except for item 6, which loaded onto a second factor with a high cross-loading onto the first factor. Communalities were very low (< .30) for two items (2, 5) and below .50 for items 6 and 7. Items 2 and 5 also showed lower loadings onto the primary factor than the other items, and item 2 had a negative secondary loading. In addition to the two-factor EFA, we also conducted a second EFA with just one factor, representing trust. The single-factor solution explained 48% of variance. Most items loaded well onto the single factor, although items 2, 5, and 6 loaded comparatively low. Communalities were below .50 for items 2, 5, 6, and 7. Factor loadings for the two EFAs are presented in Table 2. Based on the results, we concluded that a single-factor structure was most suitable for the

TXAI, and an improvement would be possible by removing four problematic items (i.e., items 2, 5, 6, and 7). The two-factor solution was not reasonable, given that only item 6 loaded positively onto a second factor, which explained little variance.

3.3 Alternative CFAs TXAI

Based on the insights from the EFAs, we ran two additional CFAs for the TXAI items. Fit indices for these CFAs can again be found in Table 1. We started with a model where item 6 was removed. Fit values for this version improved compared to the previous model, but the χ^2 test was still significant. Thus, we considered a single-factor model with the four problematic items from EFA removed (i.e., items 2, 5, 6, and 7). This modified model showed an improved fit on all criteria compared to the first model and generally indicated a good fit. Based on these CFAs, we concluded that removing the reverse-coded item 6 resulted in a substantially better model fit for the TXAI. Furthermore, removing items 2, 5, and 7 improved the fit even more.

3.4 CFA TPA - Original Models

Next, we conducted a CFA for the TPA [17], using a single-factor model as proposed by the original authors. In addition, we conducted a second CFA, working with a two-factor model distinguishing between trust and distrust, as proposed by Spain et al. [35]. Multivariate normality of the data was not given. Thus, we used the same robust maximum likelihood estimator as above. Fit indices are presented in the lower half of Table 1. Results from the single-factor CFA indicated an inadequate model fit based on all metrics. Furthermore, item 12 loaded negatively onto the factor, while all other items showed positive loadings. The two-factor CFA provided an improved but still sub-optimal fit compared to the single-factor model. Given the sub-optimal fit in both CFAs, we explored alternative models for the TPA using EFA.

3.5 EFA TPA

Requirements for EFA were given (Bartlett's test for sphericity: $\chi^2(66) = 10587.13$, $p < .001$; Kaiser-Meyer-Olkin measures > .80), but multivariate normality was violated. We thus used the same principal axis factoring and oblique rotation methods as above. Parallel analysis and a scree-plot suggested two factors. We thus conducted a two-factor EFA for all 12 items. Factor loadings and communalities are presented in Table 3. The two-factor solution explained 63% of variance, with the first factor explaining 35% and the

¹Low χ^2 value and $p > .05$ for the χ^2 test, *RMSEA* < .06, *SRMR* $\leq$.08 and $.95 \leq$ *CFI* $\leq$ 1

Table 2: Factor loadings $> .20$ for the two-factor (left) and single-factor (right) EFA of the TXAI by Hoffman et al. [13].

No.	Item	Two-Factor EFA			Single-Factor EFA	
		PA1	PA2	h2	PA1	h2
1	I am confident in the AI. I feel that it works well.	.84		.73	.86	.73
2	The outputs of the AI are very predictable.	.45	-.36	.27	.36	.13
3	The AI is very reliable. I can count on it to be correct all the time.	.80		.62	.78	.61
4	I feel safe that when I rely on the AI I will get the right answers.	.85		.72	.85	.72
5	The AI is efficient in that it works very quickly.	.52		.28	.53	.28
6	I am wary of the AI. (R)	.33	.46	.38	.42	.17
7	The AI can perform the task better than a novice human user.	.69		.47	.69	.47
8	I like using the AI for decision making.	.81		.70	.84	.70

Note: Problematic items are marked in bold. PA1/PA2 = loadings on the first/second factor; h2 = communality. Reverse-coded items are marked with (R).

second factor 28%. The two factors correlated at .18. Communalities for all items, except items 8 and 12, were above .50, with item 12 falling below .30. All items loaded well onto one of the two factors, with primary loadings all above .40 and no secondary loadings $> .30$. The first five items loaded onto one factor, while the other seven loaded onto a second factor. In addition to the two-factor EFA, we also considered a single-factor EFA. The single-factor solution had multiple issues, with most items having communalities $< .50$ or even $< .30$. The solution explained just 37% of variance. Loadings for the first five items were all $\leq .50$, and items 1, 2, and 5 exhibited loadings below .40. Based on results from the EFA, we concluded that a two-factor solution was more suitable for the scale and that removing items 8 and 12 could result in an improved fit.

3.6 Alternative CFA TPA

Based on the insights from EFA, we conducted one final CFA for the TPA, this time without items 8 and 12. We worked with a two-factor model, where items 1 through 5 were specified to load onto one factor (*distrust*), with the remaining items loading onto a second factor (*trust*). Fit indices are again reported in Table 1. Results showed a slightly improved fit compared to the two-factor model with all items, although fit indices suggested a sub-optimal model fit. However, the fit was substantially better compared to the single-factor model. We thus concluded that a two-factor model was more suitable for the scale but that further work would be needed to reach an optimal fit.

3.7 Reliability

As an indicator of the scales' reliability, we calculated two internal consistency coefficients, α [5] and ω [24]. Values for α were interpreted following DeVellis [6]. Exact values are reported in Table 4. Coefficient α values were very good for the TXAI and across all items of the TPA, as well as separated by trust and distrust. Concerning the modified versions, internal consistency improved for the TXAI when problematic items were removed, with the most favorable results for the version with four items removed (i.e., items 2, 5, 6, and 7), followed by the version with item 6 removed. For the TPA, removing the two problematic trust items (i.e., 8 and 12) resulted in slightly higher coefficients across the trust construct and for the overall scale. These results thus further supported the

psychometric quality of both trust scales, especially the modified alternatives, while providing additional evidence in favor of a two-factor solution for the TPA scale by Jian et al.. Coefficient ω values were comparable to α for all item sets.

3.8 Correlations

Finally, we calculated Pearson's product-moment correlations for the means across the scales' items to investigate the relationship between the two trust scales. Given the above results, we worked with a two-factor conceptualization for the TPA, separating items for the constructs of trust and distrust. Furthermore, we used the distrust items from Jian et al. [17] in their non-reversed form, removed item 6 from the TXAI, and items 8 and 12 from the TPA. Results showed a strong positive correlation between trust measured with the TXAI and the trust construct of the TPA ($r(1366) = .84$, 95% CI[.82, .85], $p < .001$). The mean score across items for distrust from the TPA showed weak negative correlations with the TXAI ($r(1366) = -.23$, 95% CI[-.28, -.18], $p < .001$) and with trust measured using the TPA ($r(1366) = -.22$, 95% CI[-.27, -.17], $p < .001$).

4 DISCUSSION

Given the limited evidence on the psychometric quality of the TPA scale by Jian et al. [17] in the AI context and because the TXAI scale by Hoffman et al. [13] has never been empirically validated, we set out to investigate how the two scales performed in an AI-specific setting. For this, we worked with data from an online study ($N = 1368$) where participants interacted with an algorithm introduced as an AI in the real estate valuation domain as past research did [27]. Using this data, we performed multiple analyses to investigate the validity (i.e., CFA, EFA, and correlations) and reliability (i.e., coefficients α and ω) of the two scales.

Results suggest very good psychometric quality for Hoffman et al.'s TXAI, both regarding validity and reliability. Furthermore, removing the lone reverse-coded item 6 resulted in an even better fit, while removing three additional items improved the fit even more (i.e., items 2, 5, and 7). Given the excellent fit for the version with only item 6 removed, we suggest researchers use this scale version. We recommend this version over the alternative version with four items removed for two reasons: First, while the CFA model fit improved after items 2, 5, and 7 were removed from

Table 3: Factor loadings $> .20$ for the two-factor (left) and single-factor (right) EFA of the TPA by Jian et al. [17].

No.	Item	Two-Factor EFA			Single-Factor EFA	
		PA1	PA2	h2	PA1	h2
1	The AI-system is deceptive (R)		.87	.75	.37	.14
2	The AI-system behaves in an underhanded manner (R)		.84	.68	.27	.07
3	I am suspicious of the AI-system's intent, action or, outputs (R)		.80	.70	.49	.24
4	I am wary of the AI-system (R)		.72	.59	.50	.25
5	The AI-system's actions will have a harmful or injurious outcome (R)		.79	.61	.30	.09
6	I am confident in the AI-system	.87		.78	.86	.74
7	The AI-system provides security	.77		.58	.63	.40
8	The AI-system has integrity	.68	-.20	.45	.53	.28
9	The AI-system is dependable	.79		.67	.81	.66
10	The AI-system is reliable	.84		.74	.85	.71
11	I can trust the AI-system	.86		.75	.83	.70
12	I am familiar with the AI-system	.52	-.24	.28	.37	.14

Note: Problematic items are marked in bold. PA1/PA2 = loadings on the first/second factor; h2 = communality. Reverse-coded items are marked with (R).

the scale, the version without item 6 already showed a good fit. Furthermore, the three items loaded well in the two-factor EFA, with only the communality values suggesting that the items are sub-optimal. Thus, from a psychometric perspective, only a slight improvement comes from removing these three items compared to just removing item 6. Second, removing only one item means that researchers stay closer to the original TXAI. Thus, they are likely to capture more aspects of trust and stay closer to Hoffman et al.'s original intentions while keeping higher comparability of results to the initial version of the TXAI. Excluding item 6 has an additional advantage beyond the improved model fit: no negatively worded items are used in the scale. Previous research has shown that these reversed items negatively impact survey scales' quality [1, 26, 32, 33, 36, 38]. Additionally, negatively worded items may be misinterpreted by participants and are prone to coding mistakes by researchers [29]. Ultimately, reversed items do more harm than the biases they aim to mitigate, leading Sauro and Lewis [29] to advise that HCI researchers avoid negative item wording in survey scales.

Concerning the TPA scale by Jian et al. [17], the results were not in favor of the single-factor solution proposed by the original authors. However, separating items to load onto two factors, one for trust and a second for distrust, resulted in improved values for the scale in terms of validity and reliability. Thus, our results favor

a two-factor model for the TPA, as previously suggested by Spain et al. [35] outside of an AI context, over the proposed single-factor solution. Removing two items (i.e., 8 and 12) further increased the TPA's validity and reliability. The scale thus can be used in an AI context, despite its original development context in automation, if a two-factor model is used and the suggested modifications are taken into account. Correlations among the two scales further showed that the items from the TXAI correlate closely with the items for trust from the TPA, suggesting that both scales measure the same construct (i.e., trust). In addition, given the substantial correlation between the mean values across the two scales, they deliver comparable results. Furthermore, these trust items showed comparable negative correlations to the items from the TPA measuring distrust. Thus, the TXAI's items and the trust items from the TPA appear to measure a distinct construct from the TPA's distrust items. Our results for the TPA scale align with the findings of Spain et al. [35], suggesting that trust and distrust are distinct factors and not the same underlying concept. This separation has important implications and supports the argument of disentangling trust from distrust for a simultaneous coexistence of these two distinct attitudes [21]. A more nuanced understanding of the dynamics of trust and distrust is needed in developing AI systems that foster trust and prevent distrust as separate but critical criteria for developing new systems. There are factors contributing to increases and decreases in trust, along with factors contributing to increases and decreases in distrust [21]. Researchers implementing the TPA must account for this distinction between trust and distrust. By doing so, the adapted version of the scale allows them to confidently measure both aspects in the context of XAI with increased validity compared to a single-factor conceptualization of trust.

Table 4: Coefficients α and ω for the two trust scales (including 95% confidence intervals).

Scale	α	ω
TXAI [13] (all items)	.86 [.84, .87]	.87 [.86, .88]
TXAI [13] (item 6 removed)	.87 [.86, .88]	.88 [.87, .89]
TXAI [13] (items 2, 5, 6, & 7 removed)	.90 [.89, .91]	.90 [.89, .91]
TPA [17] (all items)	.85 [.83, .86]	.81 [.77, .84]
TPA [17] (trust items)	.89 [.88, .90]	.89 [.88, .90]
TPA [17] (distrust items)	.90 [.89, .91]	.90 [.89, .91]
TPA [17] (all items, without 8 & 12)	.86 [.85, .87]	.85 [.83, .87]
TPA [17] (trust, items 8 & 12 removed)	.92 [.91, .92]	.92 [.91, .93]

4.1 Limitations and Next Steps

We investigated trust in a research study setting, which may not generalize to every real-world situation, but allowed for more control over confounding variables. Future work should explore the

observed differentiation between trust and distrust in various settings to better understand their relationship and predictive power (e.g., for behavioral measures). Furthermore, we recruited participants from the USA on Amazon's Mechanical Turk. It could be insightful to investigate the scales' measurement invariance across a wider range of contexts, or sub-groups of varying age, gender, and cultural backgrounds.

5 CONCLUSION

This paper provides initial evidence for the psychometric quality of the TPA [17] within an AI context. Furthermore, the validity and reliability of the TXAI by Hoffman et al. [13] are investigated for the first time. Results from an online study ($N = 1368$) show that the TXAI performs well within an AI context, as long as one item is removed (i.e., item 6). Concerning the TPA, results favor working with a two-factor model, differentiating between trust and distrust. Overall, our results suggest that AI researchers wanting to measure trust are best served using the single-factor TXAI, without item 6. If researchers are interested in distinguishing between trust and distrust, they are better off using a two-factor model with items 8 and 12 removed when working with the TPA. Overall, results show that AI researchers should work with the TXAI if they are only interested in measuring trust, while they should use a two-factor model of trust and distrust when working with the TPA. Not only is there a need for trust measures that meet psychometric standards, but it also seems necessary to expand the current understanding of AI trust and consider related constructs such as distrust to gain a more comprehensive understanding of the dynamic interaction between humans and AI.

REFERENCES

- [1] J. Jackson Barnette. 2000. Effects of Stem and Likert Response Option Reversals on Survey Internal Consistency: If You Feel the Need, There is a Better Alternative to Using those Negatively Worded Stems. *Educational and Psychological Measurement* 60, 3 (2000), 361–370. <https://doi.org/10.1177/00131640021970592>
- [2] Florian Brühlmann, Serge Petralito, Lena Aeschbach, and Klaus Opwis. 2020. The Quality of Data Collected Online: An Investigation of Careless Responding in a Crowdsourced Sample. *Methods in Psychology* 2 (04 2020), 100022. <https://doi.org/10.1016/j.metip.2020.100022>
- [3] Zana Bućinca, Phoebe Lin, Krzysztof Z. Gajos, and Elena L. Glassman. 2020. Proxy Tasks and Subjective Measures Can Be Misleading in Evaluating Explainable AI Systems. In *Proceedings of the 25th International Conference on Intelligent User Interfaces* (Cagliari, Italy) (IUI '20). ACM, New York, NY, USA, 454–464. <https://doi.org/10.1145/3377325.3377498>
- [4] John T Cacioppo and Gary G Berntson. 1994. Relationship Between Attitudes and Evaluative Space: A Critical Review, with Emphasis on the Separability of Positive and Negative Substrates. *Psychological Bulletin* 115, 3 (1994), 401–423. <https://doi.org/10.1037/0033-2909.115.3.401>
- [5] Lee J Cronbach. 1951. Coefficient Alpha and the Internal Structure of Tests. *Psychometrika* 16, 3 (1951), 297–334. <https://doi.org/10.1007/BF02310555>
- [6] Robert F. DeVellis. 2017. *Scale Development: Theory and Applications* (4th ed.). SAGE publications, Inc., Thousand Oaks, CA, USA.
- [7] Alice H Eagly and Shelly Chaiken. 1993. *The Psychology of Attitudes*. Harcourt Brace Jovanovich College Publishers.
- [8] Mike Furr. 2011. *Scale Construction and Psychometrics for Social and Personality Psychology*. SAGE publications, Ltd., London, UK.
- [9] Robert S. Gutzwiller, Erin K. Chiou, Scotty D. Craig, Christina M. Lewis, Glenn J. Lematta, and Chi-Ping Hsiung. 2019. Positive Bias in the "Trust in Automated Systems Survey"? An Examination of the Jian et al. (2000) Scale. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 63, 1 (2019), 217–221. <https://doi.org/10.1177/1071181319631201>
- [10] Joseph F. Hair, William C. Black, Barry J. Babin, and Rolph E. Anderson. 2010. *Multivariate Data Analysis* (7th ed.). Prentice Hall, Hoboken, NJ, USA. 785 pages.
- [11] N Henze and B Zirkler. 1990. A Class of Invariant Consistent Tests for Multivariate Normality. *Communications in statistics-Theory and Methods* 19, 10 (1990), 3595–3617.
- [12] Kevin A. Hoff and Masooda Bashir. 2015. Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust. *Human Factors* 57, 3 (2015), 407–434. <https://doi.org/10.1177/0018720814547570>
- [13] Robert R. Hoffman, Shane T. Mueller, Gary Klein, and Jordan Litman. 2023. Measures for explainable AI: Explanation goodness, user satisfaction, mental models, curiosity, trust, and human-AI performance. *Frontiers in Computer Science* 5 (2023), 15 pages. <https://doi.org/10.3389/fcomp.2023.1096257>
- [14] Kenneth D Hopkins. 1998. *Educational and Psychological Measurement and Evaluation*. Pearson, London, UK.
- [15] Matt C. Howard. 2016. A Review of Exploratory Factor Analysis Decisions and Overview of Current Practices: What We Are Doing and How Can We Improve? *International Journal of Human-Computer Interaction* 32, 1 (2016), 51–62. <https://doi.org/10.1080/10447318.2015.1087664>
- [16] Li-tze Hu and Peter M. Bentler. 1999. Cutoff Criteria for Fit Indexes in Covariance Structure Analysis: Conventional Criteria Versus new Alternatives. *Structural Equation Modeling: A Multidisciplinary Journal* 6, 1 (1999), 1–55. <https://doi.org/10.1080/10705519909540118>
- [17] Jiun-Yin Jian, Ann M. Bisantz, and Colin G. Drury. 2000. Foundations for an Empirically Determined Scale of Trust in Automated Systems. *International Journal of Cognitive Ergonomics* 4, 1 (2000), 53–71. https://doi.org/10.1207/S15327566IJCE0401_04
- [18] Elizabeth F. Juniper. 2009. Validated Questionnaires Should not be Modified. *European Respiratory Journal* 34, 5 (2009), 1015–1017. <https://doi.org/10.1183/09031936.00110209>
- [19] Markus Langer, Tim Hunsicker, Tina Feldkamp, Cornelius J. König, and Nina Grgić-Hlača. 2022. "Look! It's a Computer Program! It's an Algorithm! It's AI!": Does Terminology Affect Human Perceptions and Evaluations of Algorithmic Decision-Making Systems?. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 581, 28 pages. <https://doi.org/10.1145/3491102.3517527>
- [20] John D. Lee and Katrina A. See. 2004. Trust in Automation: Designing for Appropriate Reliance. *Human factors* 46, 1 (2004), 50–80. <https://doi.org/10.1518/hfes.46.1.50.30392>
- [21] Roy J Lewicki, Daniel J McAllister, and Robert J Bies. 1998. Trust and Distrust: New Relationships and Realities. *Academy of management Review* 23, 3 (1998), 438–458.
- [22] Kantilal Vardichand Mardia. 1970. Measures of Multivariate Skewness and Kurtosis with Applications. *Biometrika* 57, 3 (1970), 519–530. <https://doi.org/10.1093/biomet/57.3.519>
- [23] Roger C. Mayer, James H. Davis, and F. David Schoorman. 1995. An Integrative Model of Organizational Trust. *Academy of Management Review* 20, 3 (1995), 709–734. <https://doi.org/10.2307/258792>
- [24] Roderick P McDonald. 1999. *Test Theory: A Unified Treatment*. Lawrence Erlbaum Associates, Inc., Mahwah, NJ, USA.
- [25] Stephanie M. Merritt. 2011. Affective Processes in Human-Automation Interactions. *Human Factors* 53, 4 (2011), 356–370. <https://doi.org/10.1177/0018720811411912>
- [26] William J. Pilotte and Robert K. Gable. 1990. The Impact of Positive and Negative Item Stems on the Validity of a Computer Anxiety Scale. *Educational and Psychological Measurement* 50, 3 (1990), 603–610. <https://doi.org/10.1177/0013164490503016>
- [27] Forough Poursabzi-Sangdeh, Daniel G. Goldstein, Jake M. Hofman, Jennifer Wortman Wortman Vaughan, and Hanna Wallach. 2021. Manipulating and Measuring Model Interpretability. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). ACM, New York, NY, USA, Article 237, 52 pages. <https://doi.org/10.1145/3411764.3445315>
- [28] Joseph R Priester and Richard E Petty. 1996. The Gradual Threshold Model of Ambivalence: Relating the Positive and Negative Bases of Attitudes to Subjective Ambivalence. *Journal of personality and social psychology* 71, 3 (1996), 431.
- [29] Jeff Sauro and James R. Lewis. 2011. When Designing Usability Questionnaires, Does It Hurt to Be Positive?. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI '11). Association for Computing Machinery, New York, NY, USA, 2215–2224. <https://doi.org/10.1145/1978942.1979266>
- [30] Nicolas Scharowski, Sebastian AC Perrig, Nick von Felten, and Florian Brühlmann. 2022. Trust and Reliance in XAI-Distinguishing Between Attitudinal and Behavioral Measures. *CHI TRAIT Workshop* (2022), 6 pages. <https://doi.org/10.48550/arXiv.2203.12318>
- [31] Beau G Schelble, Christopher Flathmann, Matthew Scalia, Shiwen Zhou, Christopher Myers, Nathan J McNeese, Jamie Gorman, and Guo Freeman. 2022. Addressing the Spread of Trust and Distrust in Distributed Human-AI Teaming Constellations. *CHI TRAIT Workshop* (2022), 11 pages.
- [32] Neal Schmitt and Daniel M. Stuits. 1985. Factors Defined by Negatively Keyed Items: The Result of Careless Respondents? *Applied Psychological Measurement* 9, 4 (1985), 367–373. <https://doi.org/10.1177/014662168500900405>
- [33] Chester A. Schriesheim and Kenneth D. Hill. 1981. Controlling Acquiescence Response Bias by Item Reversals: The Effect on Questionnaire Validity. *Educational*

- and *Psychological Measurement* 41, 4 (1981), 1101–1114. <https://doi.org/10.1177/001316448104100420>
- [34] Randall E Schumacker and Richard G Lomax. 2010. *A beginner's guide to structural equation modeling* (3 ed.). Routledge, New York, NY, USA. <https://doi.org/10.4324/9780203851319>
- [35] Randall D. Spain, Ernesto A. Bustamante, and James P. Bliss. 2008. Towards an Empirically Developed Scale for System Trust: Take Two. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 52, 19 (2008), 1335–1339. <https://doi.org/10.1177/154193120805201907>
- [36] Thomas J Stewart and Ann W Frye. 2004. Investigating the use of Negatively Phrased Survey Items in Medical Education Settings: Common Wisdom or Common Mistake? *Academic Medicine* 79, 10 (2004), 18–20.
- [37] Takane Ueno, Yuto Sawa, Yeongdae Kim, Jacqueline Urakami, Hiroki Oura, and Katie Seaborn. 2022. Trust in Human-AI Interaction: Scoping Out Models, Measures, and Methods. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 254, 7 pages. <https://doi.org/10.1145/3491101.3519772>
- [38] Nancy Wong, Aric Rindfleisch, and James E. Burroughs. 2003. Do Reverse-Worded Items Confound Measures in Cross-Cultural Consumer Research? The Case of the Material Values Scale. *Journal of Consumer Research* 30, 1 (2003), 72–91. <https://doi.org/10.1086/374697>
- [39] Ming Yin, Jennifer Wortman Vaughan, and Hanna Wallach. 2019. Understanding The Effect of Accuracy on Trust in Machine Learning Models. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3290605.3300509>
- [40] Kun Yu, Shlomo Berkovsky, Ronnie Taib, Dan Conway, Jianlong Zhou, and Fang Chen. 2017. User Trust Dynamics: An Investigation Driven by Differences in System Performance. In *Proceedings of the 22nd international conference on intelligent user interfaces* (Limassol, Cyprus) (IUI '17). Association for Computing Machinery, New York, NY, USA, 307–317. <https://doi.org/10.1145/3025171.3025219>